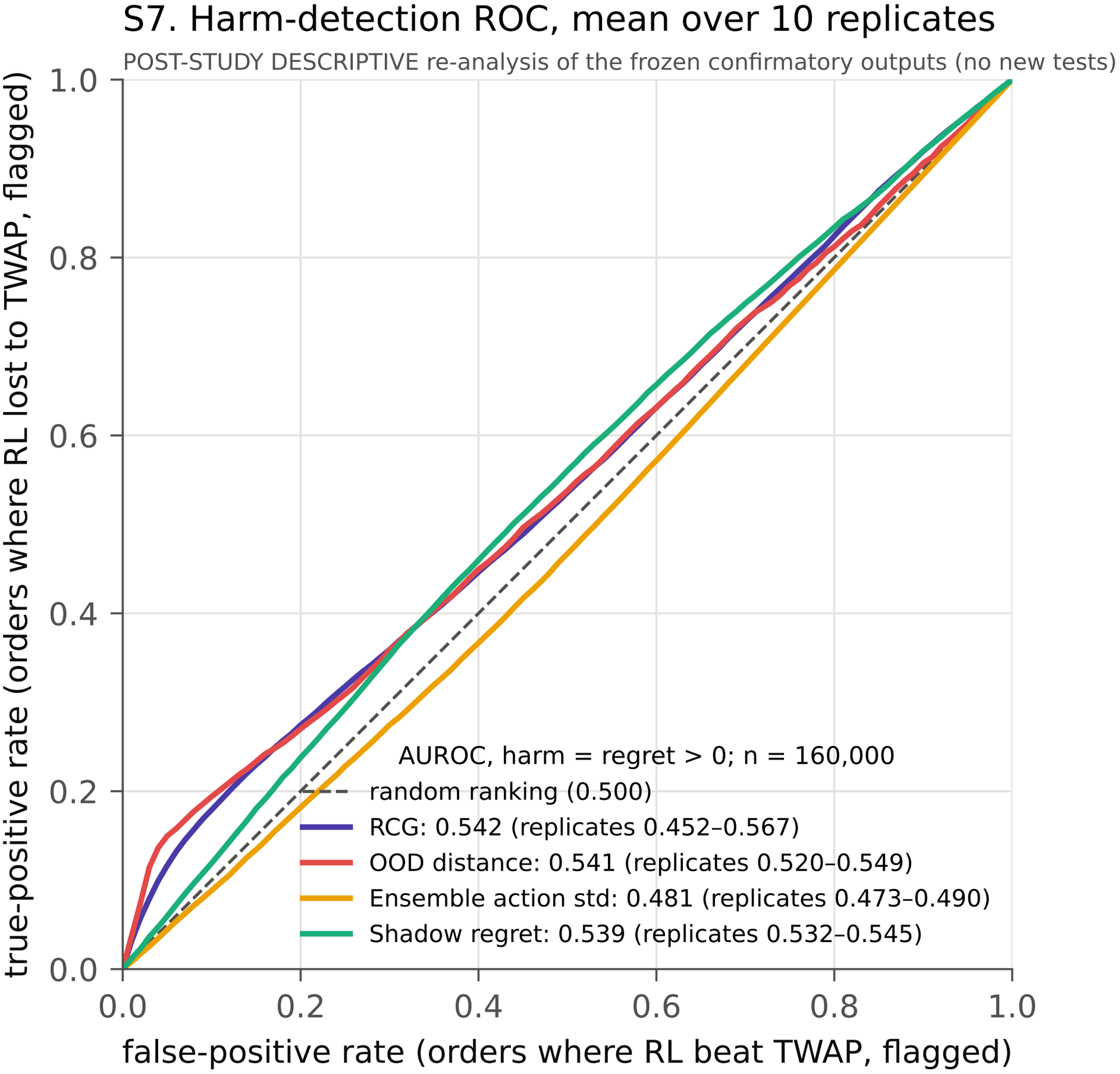
S7. Harm-detection ROC, mean over 10 replicates
POST-STUDY DESCRIPTIVE re-analysis of the frozen confirmatory outputs (no new tests)
true-positive rate (orders where RL lost to TWAP, flagged)
false-positive rate (orders where RL beat TWAP, flagged)
0.0
0.2
0.4
0.6
0.8
1.0
AUROC, harm = regret > 0; n = 160,000
random ranking (0.500)
RCG: 0.542 (replicates 0.452–0.567)
OOD distance: 0.541 (replicates 0.520–0.549)
Ensemble action std: 0.481 (replicates 0.473–0.490)
Shadow regret: 0.539 (replicates 0.532–0.545)